BLOOD

REIKA ATSUMI, MD
MARIA NOVIEL DOMINGUEZ, MD

- Function and Characteristics
- Composition of Blood
  - Plasma
  - Formed Elements
- Preventing Blood Loss

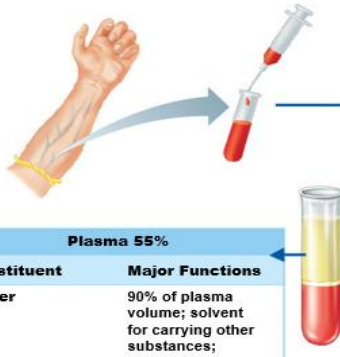
FUNCTION AND CHARACTERISTICS

- Blood transports everything that must be carried from one place to another, such as:
  - Nutrients
  - Wastes
  - Hormones
  - Body heat
- Blood characteristics
  - Sticky, opaque fluid
  - Heavier and thicker than water
  - Color range
    - Oxygen-rich blood is scarlet red
    - Oxygen-poor blood is dull red or purple
  - Metallic, salty taste
  - Blood pH is slightly alkaline, between 7.35 and 7.45
  - Blood temperature is slightly higher than body temperature, at 38ºC or 100.4ºF
- Blood volume

  About 5–6 liters, or about 6 quarts, of blood are found in a healthy adult
  Blood makes up 8 percent of body weight

COMPOSITION OF BLOOD

- Blood is the only fluid tissue, a type of connective tissue, in the human body
- Components of blood
  - Plasma (nonliving fluid matrix)
  - Formed elements (living cells)
    - ✓ Erythrocytes / RBC
    - ✓ Leukocytes / WBC
    - ✓ Thrombocytes / Platelets
  - When blood is separated:
    - ✓ Erythrocytes sink to the bottom known as the *hematocrit*
    - ✓ Buffy coat contains leukocytes and platelets (less than 1 percent of blood)
    - ✓ Buffy coat is a thin, whitish layer between the erythrocytes and plasma
    - ✓ Plasma rises to the top



Plasma 55%

Constituent	Major Functions
Water	90% of plasma volume; solvent for carrying other substances; absorbs heat
Salts (electrolytes): Sodium, Potassium, Calcium, Magnesium, Chloride, Bicarbonate	Osmotic balance, pH buffering, regulation of membrane permeability
Plasma proteins: Albumin	Osmotic balance, pH buffering
Fibrinogen	Clotting of blood
Globulins	Defense (antibodies) and lipid transport
Substances transported by blood	
Nutrients (glucose, fatty acids, amino acids, vitamins)	
Waste products of metabolism (urea, uric acid)	
Respiratory gases (O_2 and CO_2)	
Hormones (steroids and thyroid hormone are carried by plasma proteins)	

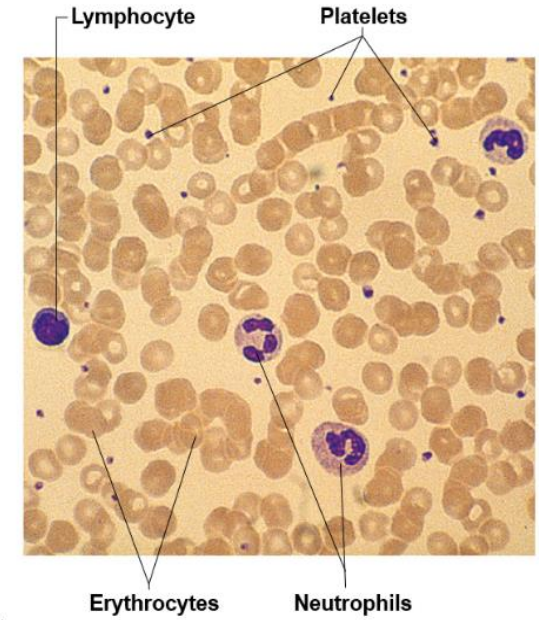
Formed elements (cells) 45%

Cell Type	Number (per mm^3 of blood)	Functions
Erythrocytes (red blood cells)	4–6 million	Transport oxygen and help transport carbon dioxide
Leukocytes (white blood cells)	4,800–10,800	Defense and immunity
Basophil		
Eosinophil		Lymphocyte
Neutrophil		Monocyte
Platelets	250,000–400,000	Blood clotting

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Plasma

- 90 percent water
- Straw-colored fluid
- Includes many dissolved substances
  - Nutrients
  - Salts (electrolytes)
  - Respiratory gases
  - Hormones
  - Plasma proteins
  - Waste products



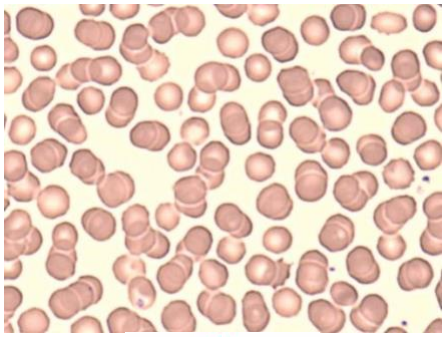
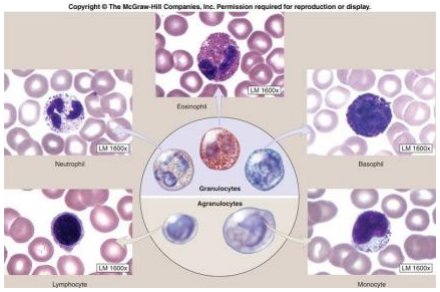
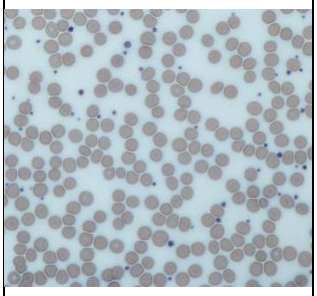
- Plasma proteins
  - Most abundant solutes in plasma
  - Most are made by the liver
  - Include:
    - Albumin—an important blood buffer and contributes to osmotic pressure
    - Clotting proteins—help to stem blood loss when a blood vessel is injured
    - Antibodies—help protect the body from pathogens
- Blood composition varies as cells exchange substances with the blood
  - Liver makes more proteins when levels drop
  - Respiratory and urinary systems restore blood pH to normal when blood becomes too acidic or alkaline
- Plasma helps distribute body heat

Formed Elements

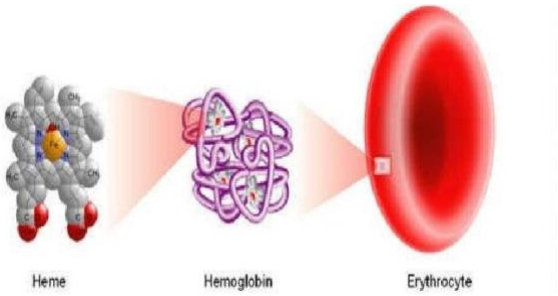
- Erythrocytes
  - Red blood cells (RBCs)
- Leukocytes
  - White blood cells (WBCs)
- Thrombocytes
  - Platelets

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FORMED ELEMENTS

❖ ERYTHROCYTES (RED BLOOD CELLS, OR RBCS)	❖ LEUKOCYTES (WHITE BLOOD CELLS, OR WBCS)	❖ PLATELETS
○ Main function is to carry oxygen ○ RBCs differ from other blood cells ▪ Anucleate (no nucleus) ▪ Contain few organelles; lack mitochondria ▪ Essentially bags of hemoglobin (Hb) ○ Shaped like biconcave discs ○ Normal count is 5 million RBCs per cubic millimeter (mm^3) of blood ○   Red Blood Cells	○ Crucial in body's defense against disease ○ Complete cells, with nucleus and organelles ○ Able to move into and out of blood vessels (diapedesis) ○ Respond to chemicals released by damaged tissues (known as *positive chemotaxis*) ○ Move by amoeboid motion ○ 4,800 to 10,800 WBCs per mm^3 of blood  ○ Leukocytosis ▪ WBC count above 11,000 cells per mm^3 of blood ▪ Generally indicates an infection ○ Leukopenia ▪ Abnormally low WBC count ▪ Commonly caused by certain drugs, such as corticosteroids and anticancer agents ○ Leukemia ▪ Bone marrow becomes cancerous ▪ Numerous immature WBC are produced ❖	○ Fragments of megakaryocytes (multinucleate cells) ○ Needed for the clotting process ○ Normal platelet count is 300,000 platelets per mm^3 of blood  ○

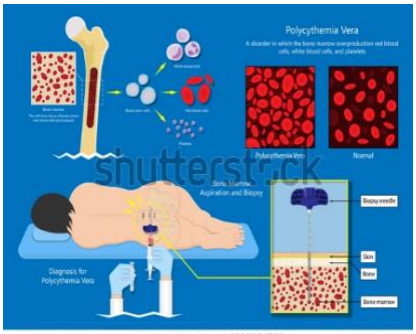
- Hemoglobin is an iron-bearing protein
  - Binds oxygen
  - Each hemoglobin molecule can bind 4 oxygen molecules
  - Each erythrocyte has 250 million hemoglobin molecules
  - Normal blood contains 12–18 g of hemoglobin per
  - 100 milliliters (ml) of blood

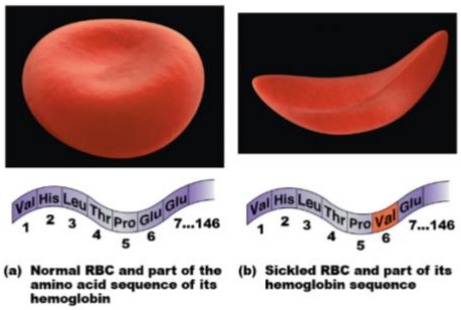


- Homeostatic imbalance of RBCs
  - Anemia is a decrease in the oxygen-carrying ability of the blood due to:
    - Lower-than-normal number of RBCs
    - Abnormal or deficient hemoglobin content in the RBCs
  - Sickle cell anemia (SCA) results from abnormally shaped hemoglobin
  - Polcythemia
    - Disorder resulting from excessive or abnormal increase of RBCs due to:

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- Bone marrow cancer (polycythemia vera)
- Life at higher altitudes (secondary polycythemia)
- Increase in RBCs slows blood flow and increases blood viscosity





(a) Normal RBC and part of the amino acid sequence of its hemoglobin

(b) Sickled RBC and part of its hemoglobin sequence

Table **10.1** **Types of Anemia**

Direct cause	Resulting from	Leading to
Decrease in RBC number	Sudden hemorrhage	Hemorrhagic anemia
	Lysis of RBCs as a result of bacterial infections	Hemolytic (he″mo-lit′ik) anemia
	Lack of vitamin B_{12} (usually due to lack of intrinsic factor required for absorption of the vitamin; intrinsic factor is formed by stomach mucosa cells)	Pernicious (per-nish′us) anemia
	Depression/destruction of bone marrow by cancer, radiation, or certain medications	Aplastic anemia
Inadequate hemoglobin content in RBCs	Lack of iron in diet or slow/prolonged bleeding (such as heavy menstrual flow or bleeding ulcer), which depletes iron reserves needed to make hemoglobin; RBCs are small and pale because they lack hemoglobin	Iron-deficiency anemia
Abnormal hemoglobin in RBCs	Genetic defect leads to abnormal hemoglobin, which becomes sharp and sickle-shaped under conditions of increased oxygen use by body; occurs mainly in people of African descent	Sickle cell anemia

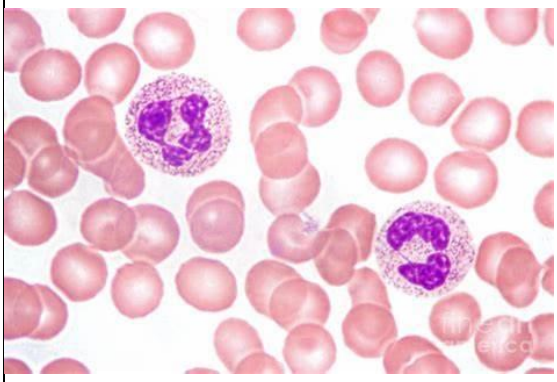
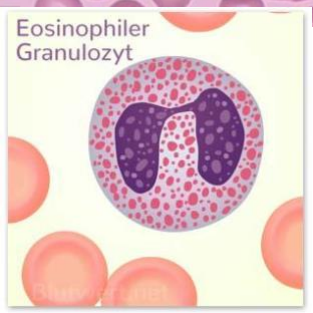
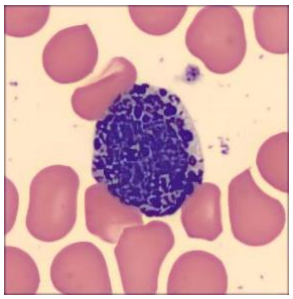
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❖ TYPES OF LEUKOCYTES	
Granulocytes	Agranulocytes
▪ Granules in their cytoplasm can be stained ▪ Possess lobed nuclei ▪ Include ✓ Neutrophils ✓ Eosinophils ✓ Basophils	▪ Lack visible cytoplasmic granules ▪ Nuclei are spherical, oval, or kidney-shaped ▪ Include ✓ Lymphocytes ✓ Monocytes

❖ List of the WBCs, from most to least abundant

- Neutrophils
- Lymphocytes
- Monocytes
- Eosinophils
- Basophils

Easy way to remember this list
Never
Let
Monkeys
Eat
Bananas

FORMED ELEMENTS: GRANULOCYTES		
○ Neutrophills	○ Eosinophils	○ Basophils
○ Most numerous WBC ○ Multilobed nucleus ○ Cytoplasm stains pink and contains fine granules ○ Function as phagocytes at active sites of infection ○ Numbers increase during infection ○ 3,000–7,000 neutrophils per mm^3 of blood (40–70 percent of WBCs)	○ Nucleus stains blue-red ○ Brick-red cytoplasmic granules ○ Function is to kill parasitic worms and play a role in allergy attacks ○ 100–400 eosinophils per mm^3 of blood (1–4 percent of WBCs)	○ Rarest of the WBCs ○ Large histamine-containing granules that stain dark blue ○ Contain heparin (anticoagulant) ○ 20–50 basophils per mm^3 of blood (0–1 percent of WBCs)
	 	

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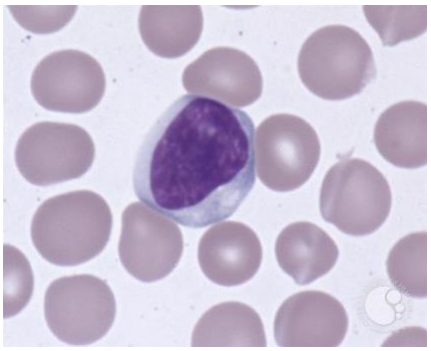
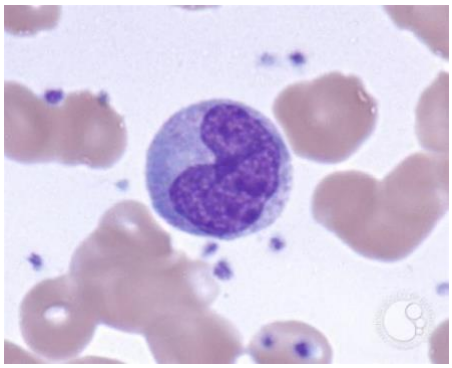
FORMED ELEMENTS: AGRANULOCYTES	
○ Lymphocytes	○ Monocytes
○ Large, dark purple nucleus ○ Slightly larger than RBCs ○ Reside in lymphatic tissues ○ Play a role in immune response ○ 1,500–3,000 lymphocytes per mm^3 of blood (20–45 percent of WBCs)  ○	○ Largest of the white blood cells ○ Distinctive U- or kidney-shaped nucleus ○ Function as macrophages when they migrate into tissues ○ Important in fighting chronic infection ○ 100–700 monocytes per mm^3 of blood (4–8 percent of WBCs) ○ 

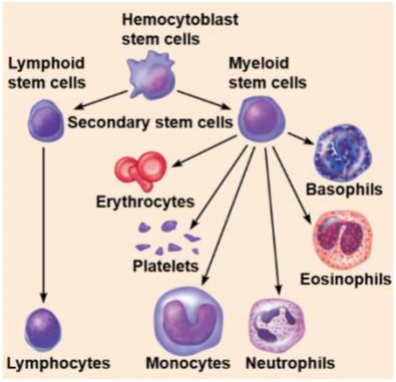
Table **10.2** **Characteristics of Formed Elements of the Blood** (*continued*)

Cell type	Occurrence in blood (cells per mm^3)	Cell anatomy*	Function
Leukocytes (white blood cells)	4,800–10,800		
Granulocytes			
• Neutrophils 	3,000–7,000 (40–70% of WBCs)	Cytoplasm stains pale pink and contains fine granules, which are difficult to see; deep purple nucleus consists of three to seven lobes connected by thin strands of nucleoplasm	Active phagocytes; number increases rapidly during short-term or acute infections
• Eosinophils 	100–400 (1–4% of WBCs)	Red coarse cytoplasmic granules; figure-8 or bilobed nucleus stains blue-red	Kill parasitic worms by deluging them with digestive enzymes; play a complex role in allergy attacks
• Basophils 	20–50 (0–1% of WBCs)	Cytoplasm has a few large blue-purple granules; U- or S-shaped nucleus with constrictions, stains dark blue	Release histamine (vasodilator chemical) at sites of inflammation; contain heparin, an anticoagulant
Erythrocytes (red blood cells) 	4–6 million	Salmon-colored biconcave disks; anucleate; literally, sacs of hemoglobin; most organelles have been ejected	Transport oxygen bound to hemoglobin molecules; also transport small amount of carbon dioxide

*Appearance when stained with Wright's stain.

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PREVENTING BLOOD LOSS



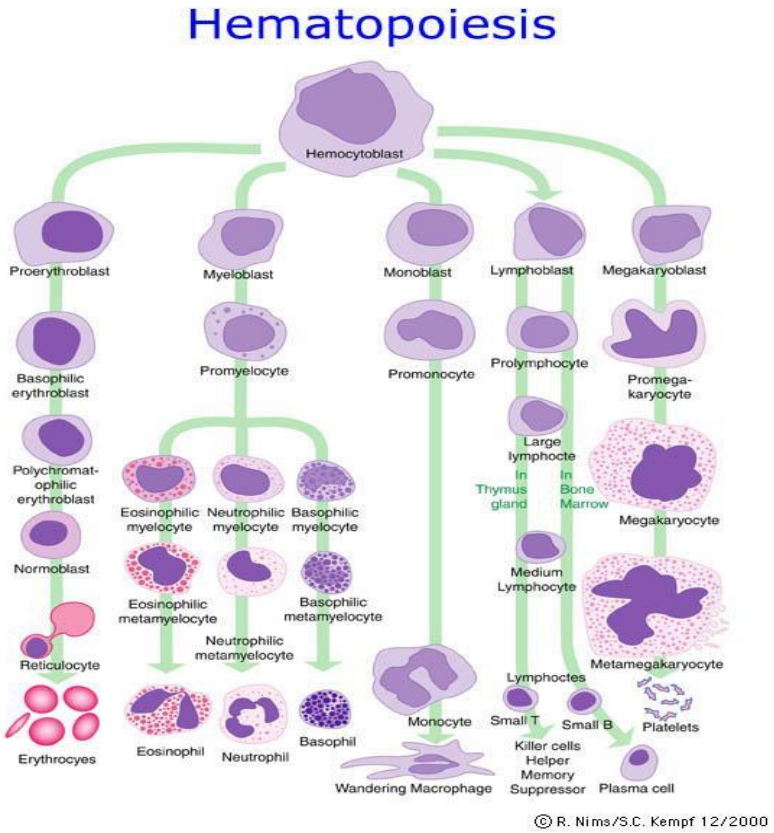
- Hematopoiesis is the process of blood cell formation
- Occurs in red bone marrow (myeloid tissue)
- All blood cells are derived from a common stem cell (hemocytoblast)
- Hemocytoblasts form two types of descendants
  - Lymphoid stem cell, which produces lymphocytes
  - Myeloid stem cell, which can produce all other formed elements

Formation of Red Blood Cells

- Since RBCs are anucleate, they are unable to divide, grow, or synthesize proteins
- RBCs wear out in 100 to 120 days
- When worn out, RBCs are eliminated by phagocytes in the spleen or liver
- Lost cells are replaced by division of hemocytoblasts in the red bone marrow
- Rate of RBC production is controlled by a hormone called *erythropoietin*
- Kidneys produce most erythropoietin as a response to reduced oxygen levels in the blood
- Homeostasis is maintained by negative feedback from blood oxygen levels

Formation of White Blood Cells and Platelets

- WBC and platelet production is controlled by hormones
  - Colony stimulating factors (CSFs) and interleukins prompt bone marrow to generate leukocytes
  - Thrombopoietin stimulates production of platelets from megakaryocytes



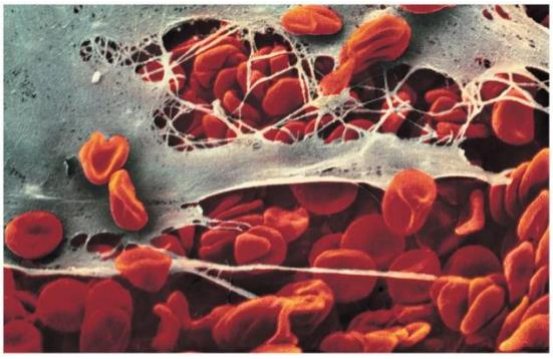
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Hemostasis

- Hemostasis is the process of stopping the bleeding that results from a break in a blood vessel
  - Step 1: vascular spasms
- Immediate response to blood vessel injury
- Vasoconstriction causes blood vessel to spasm
- Spasms narrow the blood vessel, decreasing blood loss
  - Step 2: platelet plug formation
  - Collagen fibers are exposed by a break in a blood vessel
  - Platelets become "sticky" and cling to fibers
  - Anchored platelets release chemicals to attract more platelets
  - Platelets pile up to form a platelet plug (white thrombus)
  - Step 3: coagulation
- Injured tissues release tissue factor (TF)
- PF_3 (a phospholipid) interacts with TF, blood protein clotting factors, and calcium ions to trigger a clotting cascade
- Prothrombin activator converts prothrombin to thrombin (an enzyme)
- Thrombin joins fibrinogen proteins into hairlike molecules of insoluble fibrin
- Fibrin forms a meshwork (the basis for a clot)
- Within the hour, serum is squeezed from the clot as it retracts
- Serum is plasma minus clotting proteins

Hemostasis

- Blood usually clots within 3 to 6 minutes
- The clot remains as endothelium regenerates
- The clot is broken down after tissue repair



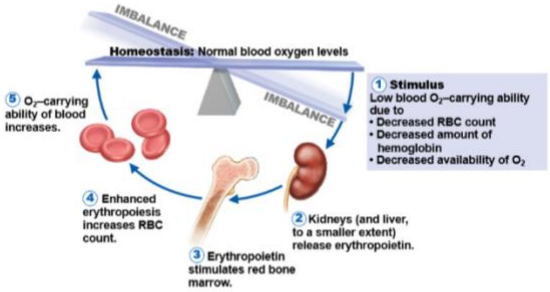
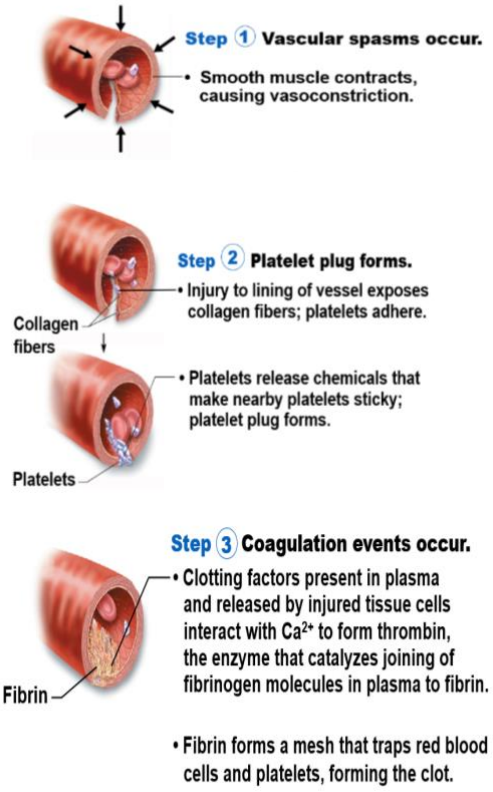


FIGURE 10.4 MECHANISM FOR REGULATING THE RATE OF RBC PRODUCTION.



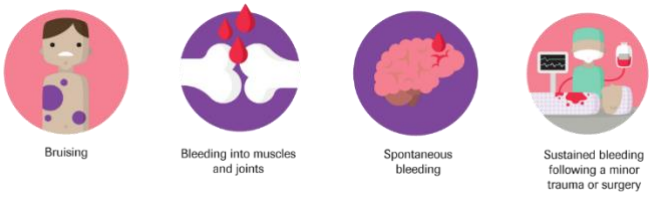
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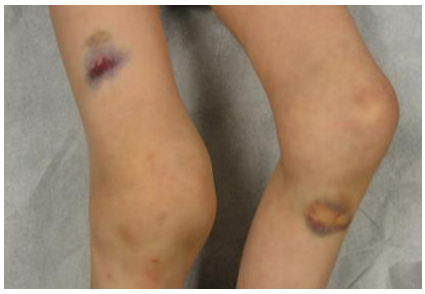
Disorders of Hemostasis



- Undesirable clotting
  - Thrombus
    - A clot in an unbroken blood vessel
    - Can be deadly in areas such as the lungs
  - Embolus
    - A thrombus that breaks away and floats freely in the bloodstream
    - Can later clog vessels in critical areas such as the brain
- Bleeding disorders
  - Thrombocytopenia
    - Insufficient number of circulating platelets
    - Arises from any condition that suppresses the bone marrow
    - Even normal movements can cause bleeding from small blood vessels that require platelets for clotting
    - Evidenced by petechiae (small purplish blotches on the skin)
  - Hemophilia
    - Hereditary bleeding disorder
    - Normal clotting factors are missing
    - Minor tissue damage can cause life-threatening prolonged bleeding







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Blood Groups and Transfusions

- Large losses of blood have serious consequences
  - Loss of 15 to 30 percent causes weakness
  - Loss of over 30 percent causes shock, which can be fatal
- Blood transfusions are given for substantial blood loss, to treat severe anemia, or for thrombocytopenia

Human Blood Groups

- Blood contains genetically determined proteins known as *antigens*
- Antigens are substances that the body recognizes as foreign and that the immune system may attack
  - Most antigens are foreign proteins
  - We tolerate our own "self" antigens
- Antibodies are the "recognizers" that bind foreign antigens
- Blood is "typed" by using antibodies that will cause blood with certain proteins to clump (agglutination) and lyse
- ABO blood group
  - Blood types are based on the presence or absence of two antigens
    - Type A
    - Type B

RELATIONSHIPS BETWEEN BLOOD TYPES AND ANTIBODIES

Blood Type	Antigens on Red Blood Cell	Can Donate Blood To	Antibodies in Cerum	Can Recieve Blood From
A	A	A, AB	Anti-B	A, O
B	B	B, AB	Anti-A	B, O
AB	A and B	AB	None	AB, O
O	None	A, B, AB, O	Anti-A and Anti-B	O

Table 10.3 ABO Blood Groups (*continued*)

Blood group	RBC antigens (agglutinogens)	Illustration	Plasma antibodies (agglutinins)	Blood that can be received	Frequency (% of U.S. population) White	Black	Asian	Native American
A	A	Anti-B, A	Anti-B (b)	A, O	40	27	28	16
O	None	Anti-B, Anti-A	Anti-A (a) Anti-B (b)	O "Universal donor"	45	49	40	79

Table 10.3 ABO Blood Groups

Blood group	RBC antigens (agglutinogens)	Illustration	Plasma antibodies (agglutinins)	Blood that can be received	Frequency (% of U.S. population) White	Black	Asian	Native American
AB	A, B	A, B	None	A, B, AB, O "Universal recipient"	4	4	5	<1
B	B	Anti-A, B	Anti-A (a)	B, O	11	20	27	4

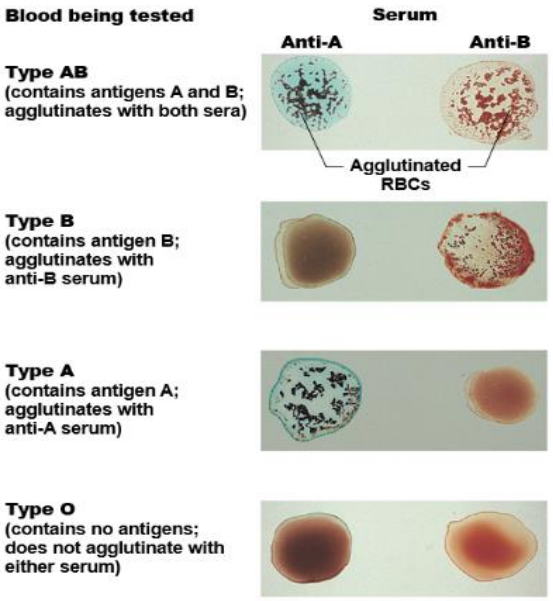
- ❖ Rh blood group
  - Named for the eight Rh antigens (agglutinogen D)
  - Most Americans are Rh^+ (Rh-positive), meaning they carry the Rh antigen
  - If an Rh^-(Rh-negative) person receives Rh+ blood:
    - The immune system becomes sensitized and begins producing antibodies; hemolysis does not occur, because as it takes time to produce antibodies
    - Second, and subsequent, transfusions involve antibodies attacking donor's Rh^+ RBCs, and hemolysis occurs (rupture of RBCs)
- Rh-related problem during pregnancy
  - Danger occurs only when the mother is Rh^-, the father is Rh^+, and the child inherits the Rh^+ factor
  - RhoGAM shot can prevent buildup of anti-Rh+ antibodies in mother's blood

- The mismatch of an Rh^- mother carrying an Rh^+ baby can cause problems for the unborn child
  - The first pregnancy usually proceeds without problems; the immune system is sensitized after the first pregnancy
  - In a second pregnancy, the mother's immune system produces antibodies to attack the Rh^+ blood (hemolytic disease of the newborn)

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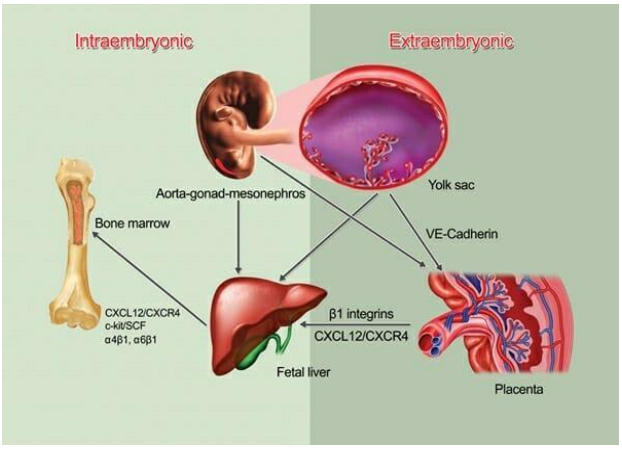
Blood Typing

- Blood samples are mixed with anti-A and anti-B serum
- Agglutination or the lack of agglutination leads to identification of blood type
- Typing for ABO and Rh factors is done in the same manner
- Cross matching—testing for agglutination of donor RBCs by the recipient's serum, and vice versa



Developmental Aspects of Blood

- Sites of blood cell formation
  - The fetal liver and spleen are early sites of blood cell formation
  - Bone marrow takes over hematopoiesis by the seventh month
  - Congenital blood defects include various types of hemolytic anemias and hemophilia
- Incompatibility between maternal and fetal blood can result in fetal cyanosis, resulting from destruction of fetal blood cells
- Fetal hemoglobin differs from hemoglobin produced after birth
- Physiologic jaundice occurs in infants when the liver cannot rid the body of hemoglobin breakdown products fast enough
- Leukemias are most common in the very young and very old
  - Older adults are also at risk for anemia and clotting disorders



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